Centrality in Collaboration: Community Detection for Oncology Researchers

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Background

- Cancer patients who do not respond to standard treatments often become candidates for clinical trials.
- It is known that the collaboration networks between oncologists is a primary factor for further engagement in subsequent trial enrollment by participants.
- Social network analysis (SNA) and community detection algorithms can be used to explore collaboration patterns.

Objectives

- The data consists of 2970 patients in 515 clinical trials at UHN's Princess Margaret Cancer Center spanning the time of January 2016 to December 2018.
- Among 389 patients enrolled in more than one of 288 trials, we aim to identify collaboration networks based on the intervention type studied in the clinical trials (Fig. 1)

Methods

- We applied the Girvan-Newman and Louvain algorithms and compared them to an author-developed algorithm that utilizes degree centrality and directionality in participant enrollments.
- Community detection was assessed by superimposing sociograms and visualizing the contrast between the in-degree/out-degree distribution.

Results

- Application of Girvan-Newman (Fig. 2), which incorporates edge betweenness but not degree centrality (Fig. 4(a)), and Louvain (Fig. 3), which does not consider directionality (Fig. 4(b)), did not yield informative structure for oncology collaboration networks.
- The algorithm developed by the authors (Fig. 5) found intuitively descriptive communities that were supported by the sociograms and visualized in-degree/out-degree distribution (Fig. 6)

Conclusions

- SNA approaches utilizing a combination of degree centrality and edge betweenness can interpret oncology collaboration networks.
- With this perspective, it is possible to delve deeper into patient engagement and improve the organizational design of clinical trial referrals.

References

1. Rostami et al., 2023, *IEEE*, 10.1109/ACCESS.2023.3260652
2. Girvan et al., 2002, *PNAS*, 10.1073/pnas.122653799
3. Blondel et al., 2008, *JSTAT*, 10.1088/1742-5468/2008/10/p10008

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Referral Network

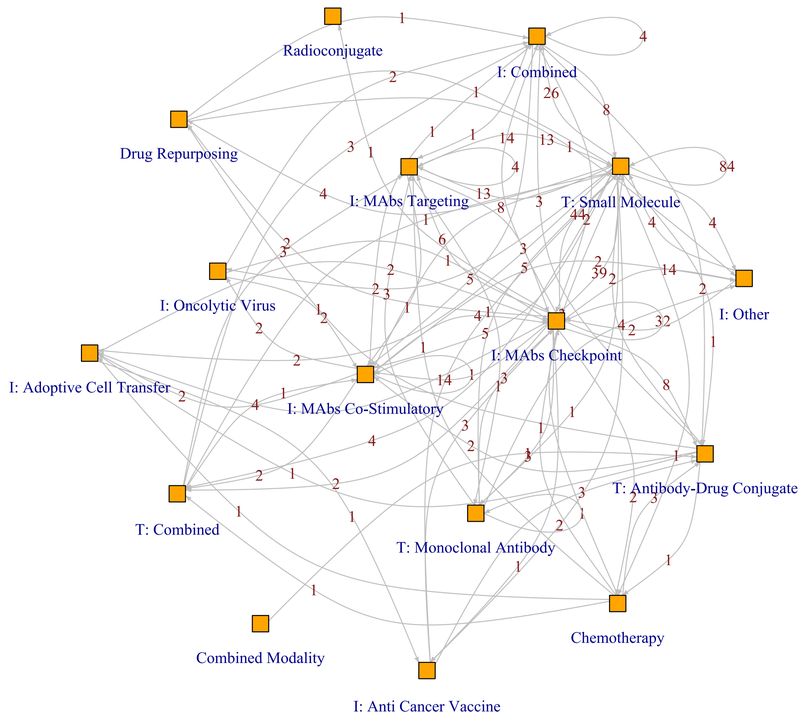


Figure 1: Patient Referral Network Based on Intervention Type Studied in Clinical Trials at UHN's Princess Margaret Cancer Center. Trials are classified to one of 16 intervention types.

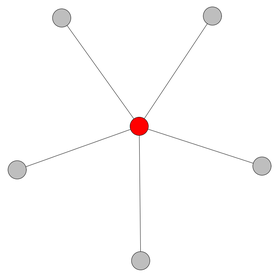


Figure 4(a): Degree Centrality, A.K.A. the number of patient referrals intervention has given/received.

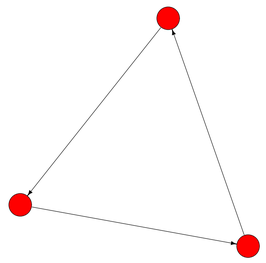


Figure 4(b): Directionality in relationships, A.K.A. the number of patients referred to/referred from an intervention type.

Pre-existing Methods

Figure 2: Girvan-Newman communities

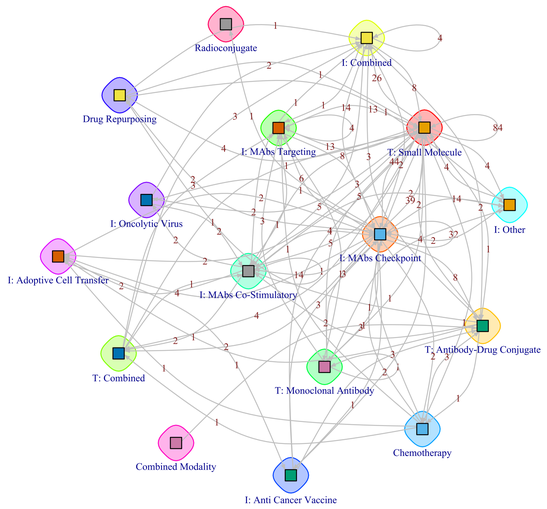
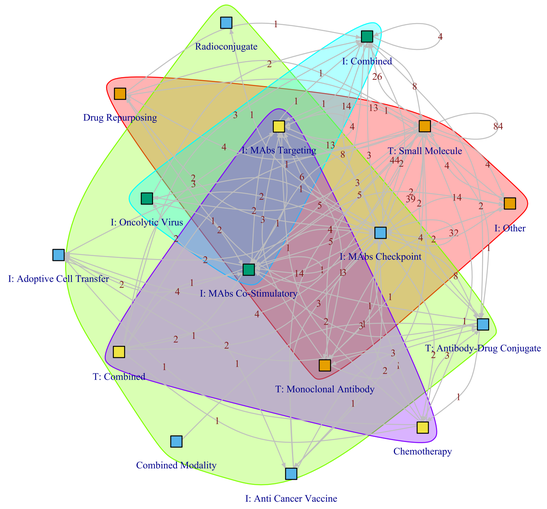


Figure 3: Louvain communities



Author Developed Method

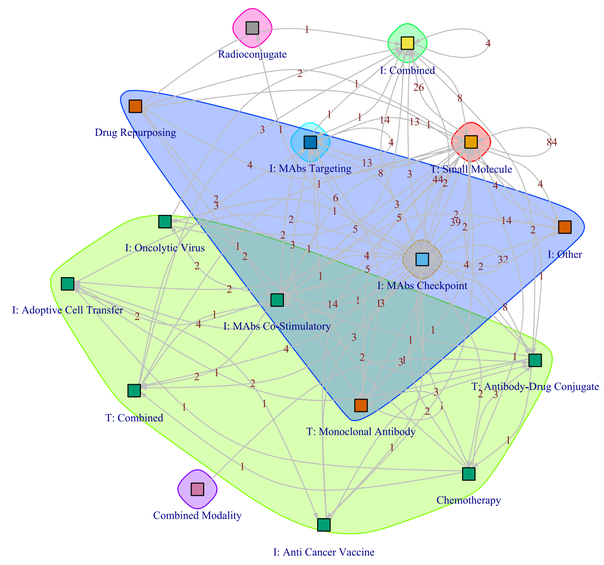


Figure 5: Author-developed algorithm communities

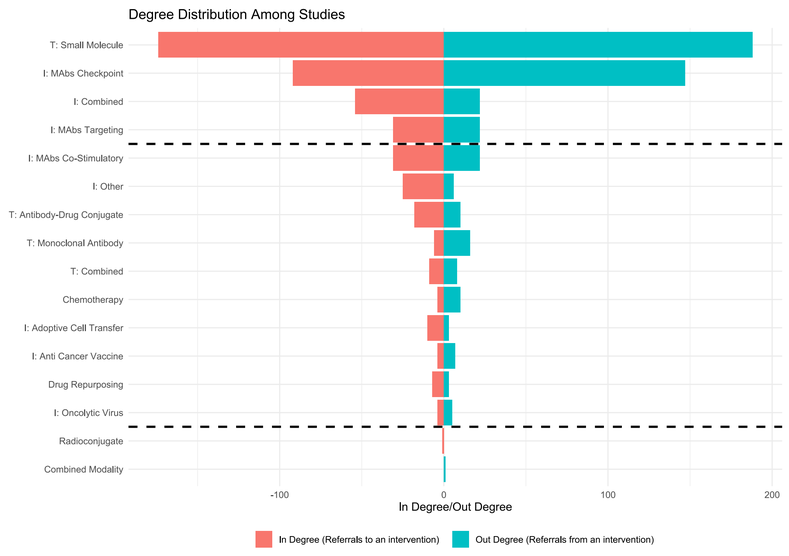


Figure 6: In-degree/out-degree distribution. The author developed algorithm's communities can be reflected in this distribution offering intuitively descriptive communities.

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